

Physics lab 3

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1 Problem 1

We first note that the probability of finding the particle at one of the walls in the box is:

$$P = |\psi(0)|^2 = |\psi(1)|^2 = 0,$$

therefore we have that $\Delta = |\psi(1)| = 0$. We used $N = [10, 100, 1000, 10000]$ mesh points and calculated the corresponding Δ value for each $j \in N$, when the ground state energy was $E = 0.5\pi^2$. We then plotted $\log(\Delta)$ vs $\log(N)$ and obtained the following figure: The red dotted line represents the theoretical slope of the error and the blue slope represents the Δ vs N in a loglog plot. The theoretical slope of $\Delta = \text{constant}/N^2$ in a loglog plot would approximately be ≈ -2 , because $\log(\Delta) = \log(k) - 2\log(N)$. Which implies,

$$\frac{\log(\Delta)}{\log(N)} = \frac{k}{\log(N)} - 2.$$

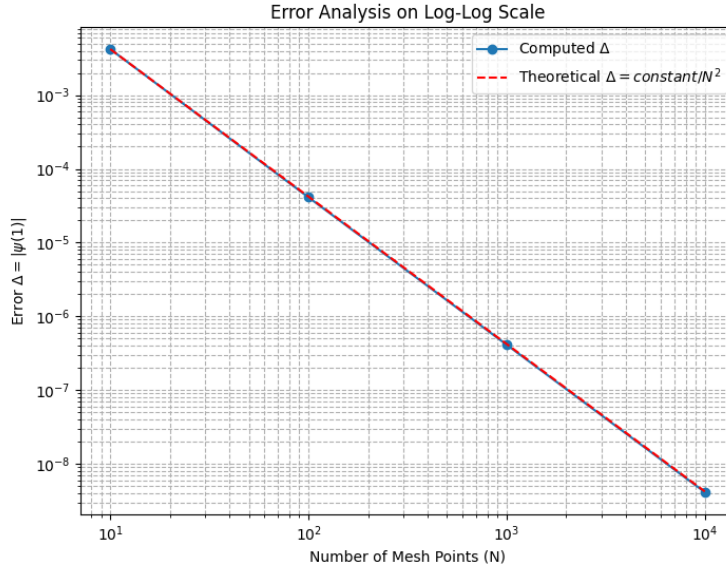


Figure 1: LogLog plot

The plot showed as expected the accuracy Δ improved as $N \rightarrow \infty$ and matched the theoretical slope.

2 Problem 2

We used the binary search algorithm and shoot method with a tolerance of 10^{-9} , $N = 10000$ mesh points, the initial guess $E = 0.4\pi^2$ and upper bound $E = 20\pi^2$ to calculate the energy eigenvalues for $n = 1, 2, 3, 4$ and its corresponding eigenstate. The resulting output gave the following eigenvalue

energies for each n :

$$\begin{aligned} n &= 1 \text{ with } E = 4.935 \\ n &= 2 \text{ with } E = 19.739 \\ n &= 3 \text{ with } E = 44.413 \\ n &= 4 \text{ with } E = 78.957 \end{aligned}$$

As expected the $\psi(1) \approx 0$ for each $n = 1, 2, 3, 4$. We also see that the oscillation of the wavefunction ψ increases with larger n , which is expected since by Broglies relation we have

$$\lambda = \frac{h}{p}.$$

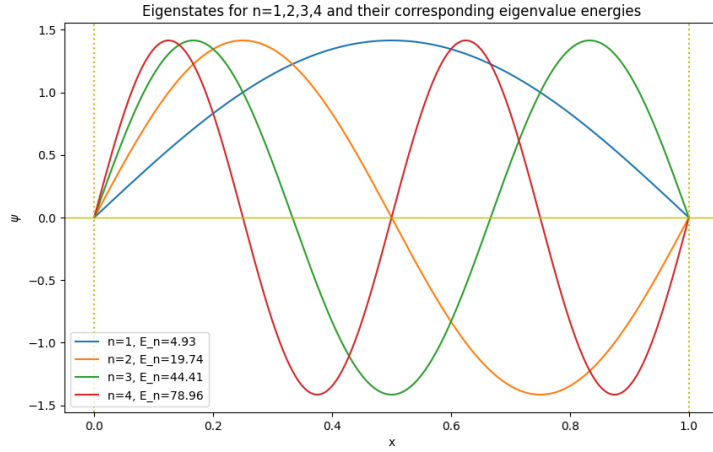


Figure 2: Eigenstates and eigenvalue energies for $n=1,2,3,4$

3 Problem 3

In this case the potential energy is given by $V(x) = \frac{1}{2}x^2$ which gives us the time independent **Schrödinger equation** as:

$$-\frac{1}{2} \frac{d^2\psi}{dx^2} + \frac{1}{2}x^2\psi = E\psi \quad (1)$$

(1) can be solved using Hermite polynomials H : which yields:

$$\psi_n(x) = A_n e^{-x^2/2} H_n(x)$$

where A_n is a normalization constant. The energy eigenvalues then become $E_n = n + 1/2$. We solved Schrödinger equation numerically (integration numerically) using $N = 1000$ mesh points with $dx = \frac{5}{N}$ and calculated the energy eigenvalues for $n = 0, 1, 2, 3$ and their corresponding eigenstates. The energy eigenvalues corresponding to n were,

$$\begin{aligned} n &= 0, E_n = 0.5 \\ n &= 1, E_n = 1.5 \\ n &= 2, E_n = 2.5 \\ n &= 3, E_n = 3.5 \end{aligned}$$

They match the exact values of $E_n = n + 1/2$. Figure 3 displays the plot for the eigenstates $\psi(x)$. We can see that the condition that $\psi(0) = 1, \psi'(0) = 0$ for n even is satisfied by looking at the slope of $\psi = 1$ (the black dotted line). Similarly we see that the condition $\psi(0) = 0, \psi'(0) = 1$ for n odd is also satisfied by looking at the slope of $\psi = x$ (black dotted line). We see that for $n = 0$ in which the Hermite polynomial is $H = 1$ we have the Gaussian distribution on $x : [0, \infty)$. In all cases for n however $\psi(x) \rightarrow 0$ as $x \rightarrow 5$.

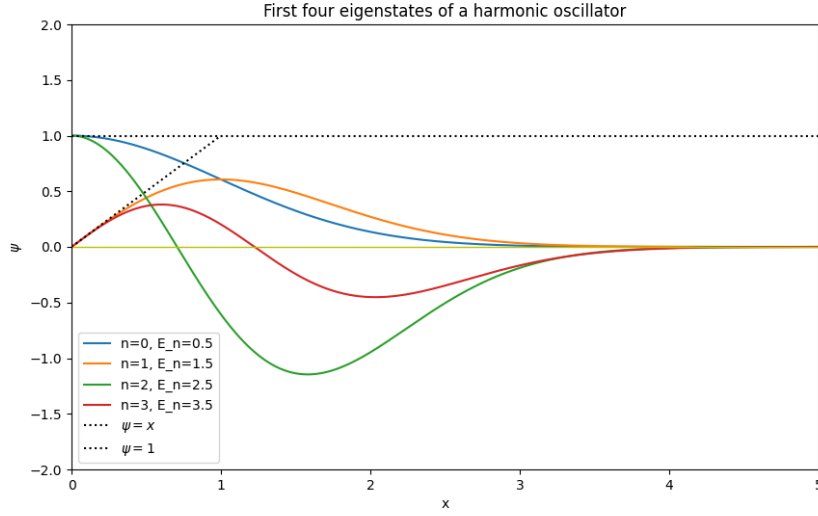


Figure 3: Eigenstates and eigenvalue energies for $n=0,1,2,3$

4 Problem 4

In this problem, we have an anharmonic potential defined as $V(x) = \frac{x^2}{2} + \frac{x^4}{2}$. We implemented the perturbation method, where we calculated the following: 1. Unperturbed Hamiltonian H_0 :

$$H_0 = -\frac{1}{2} \frac{d^2}{dx^2} + \frac{1}{2} x^2$$

Solving for the eigenvalues and eigenstates:

$$H_0 \psi_n^{(0)} = E_n^{(0)} \psi_n^{(0)}$$

2. Perturbation H_1 :

$$H_1 = \frac{x^4}{2}$$

First-order energy correction:

$$E_n^{(1)} = \langle \psi_n^{(0)} | H_1 | \psi_n^{(0)} \rangle$$

3. Corrected Eigenvalues:

$$E_n \approx E_n^{(0)} + E_n^{(1)}$$

Finally, we plotted the eigenstates computationally. The numerical solver gives the exact numerical eigenvalues:

$$n = 0, E_0 = 0.696$$

$$n = 1, E_1 = 2.324$$

$$n = 2, E_2 = 4.328$$

$$n = 3, E_3 = 6.578$$

The following figures display the results for $n = 0, 1, 2, 3$.

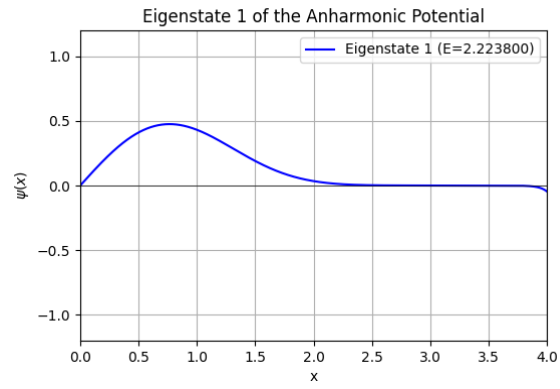


Figure 4: First eigenstate

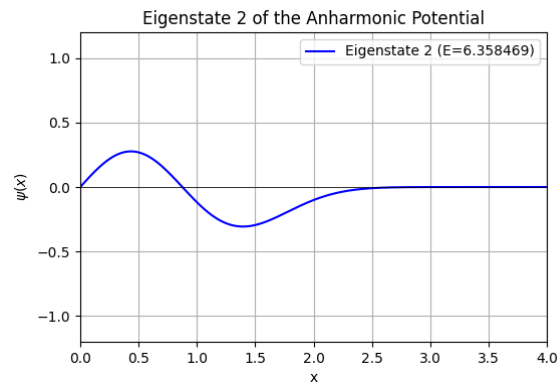


Figure 5: Second eigenstate

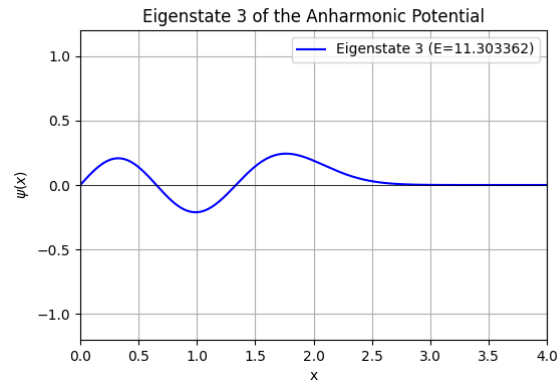


Figure 6: Third eigenstate

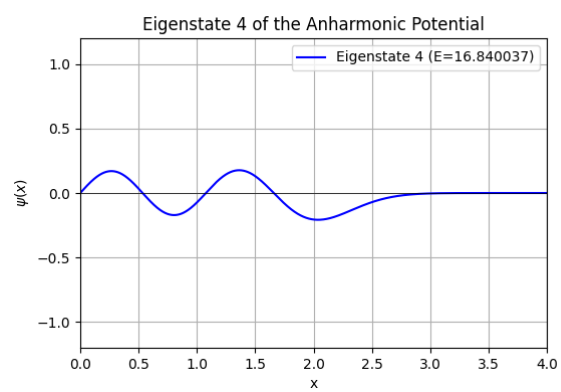


Figure 7: Fourth eigenstate